

Introduction to MapReduce

Mahmoud Parsian

Ph.D. in Computer Science

What Is MapReduce?

- A **programming model** or abstraction — **not** a programming language.
- A novel way of thinking about designing a solution to certain big data problems.
- It enables you to:
 - i. Partition data into small chunks (**partitions**)
 - ii. Execute tasks in parallel via **Mappers** and **Reducers**

Why MapReduce?

- We have access to huge volumes of data — Google processed ~24 petabytes/day in 2009; Facebook, ~60 petabytes/day in 2020.
- A single machine can't serve all of it — you need a distributed system (**cluster computing**) to store and process it in parallel.
- But parallel programming is hard: concurrency/threading, inter-node communication, scaling to more machines, machine/disk failures...

MapReduce uses cluster computing to make parallelism tractable.

A Pipe Dream, and Why It Almost Works

- Computational resources are cheap and easy to get (Amazon EC2, Microsoft Azure, ...) — so why isn't parallel programming easy too?
- Code running on one CPU is simple; on two, a headache; on four, a nightmare.
- The pipe dream: write code imagining a single CPU, and let someone else run it across thousands of machines — freeing the programmer from the distributed-systems details.
- **MapReduce is what makes that pipe dream mostly true.**

A Brief History: LISP (1958)

`map()` and `reduce()` aren't MapReduce inventions — they're functional-programming primitives from **LISP** (John McCarthy, MIT, 1960: *"Recursive Functions of Symbolic Expression and Their Computation by Machine"*):

```
map(mf, [a1, a2, ..., an])    -> [b1, b2, ..., bn]
reduce(rf, [b1, b2, ..., bn]) -> c
```

`map()` applies a function to every element of a list; `reduce()` combines a list of values into one, using another function. `map()`'s output (a list) composes directly into `reduce()`'s input.

The Analogy

1. Break a large problem into small pieces.
2. Write `mf` to solve *one* piece.
3. Run `map()` to apply `mf` to every piece, in parallel — producing nuggets of partial solutions.
4. Write `rf` to combine the nuggets.
5. Run `reduce()` to apply `rf` and produce the complete solution.

Example: a 1TB file, split into 100,000 chunks — `mf` counts lines in one chunk, `rf` adds the counts together into a final total.

MapReduce, Reintroduced

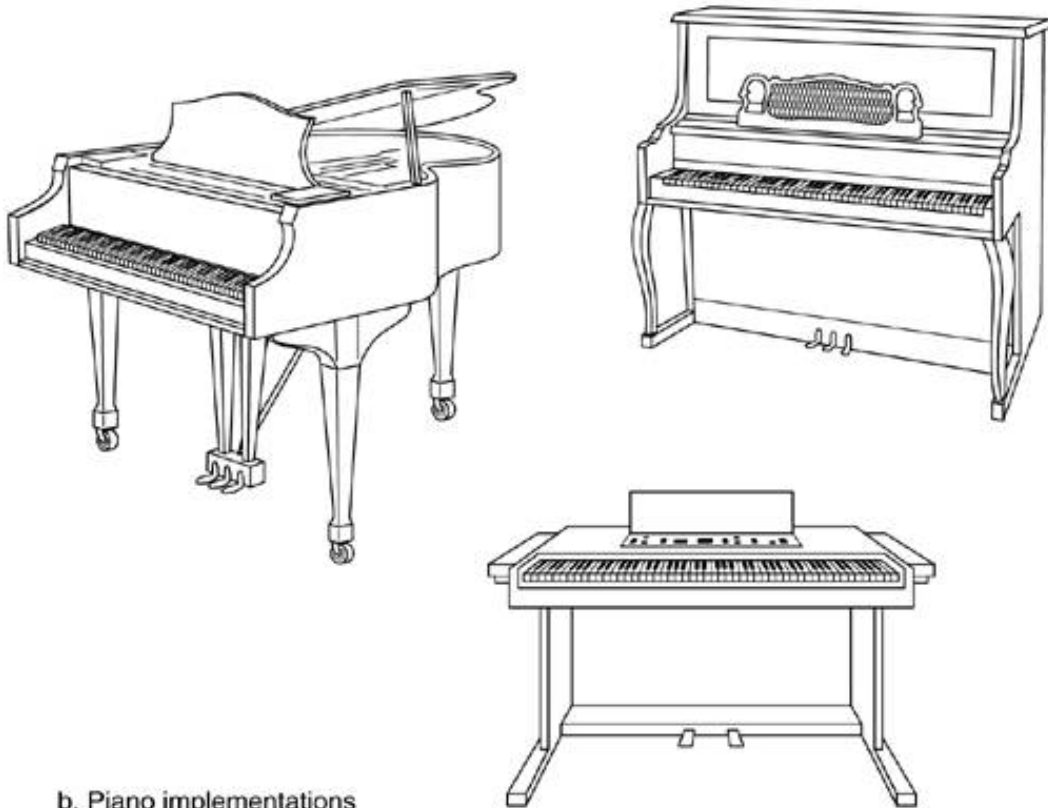
- **Google** created the awareness by publishing a paper:
"MapReduce: Simplified Data Processing on Large Clusters" —
Jeffrey Dean & Sanjay Ghemawat, OSDI'04 (Dec. 2004).
- **Apache Hadoop** made it a sensation — an open-source MapReduce implementation based on Google's paper.
- **Apache Spark** implements a superset of MapReduce (in-memory, much faster).

Google's original problem: indexing billions of web documents a day — solved by running MapReduce across 100s or 1000s of commodity servers.

MapReduce: Model vs. Implementation



a. Piano keyboard architecture



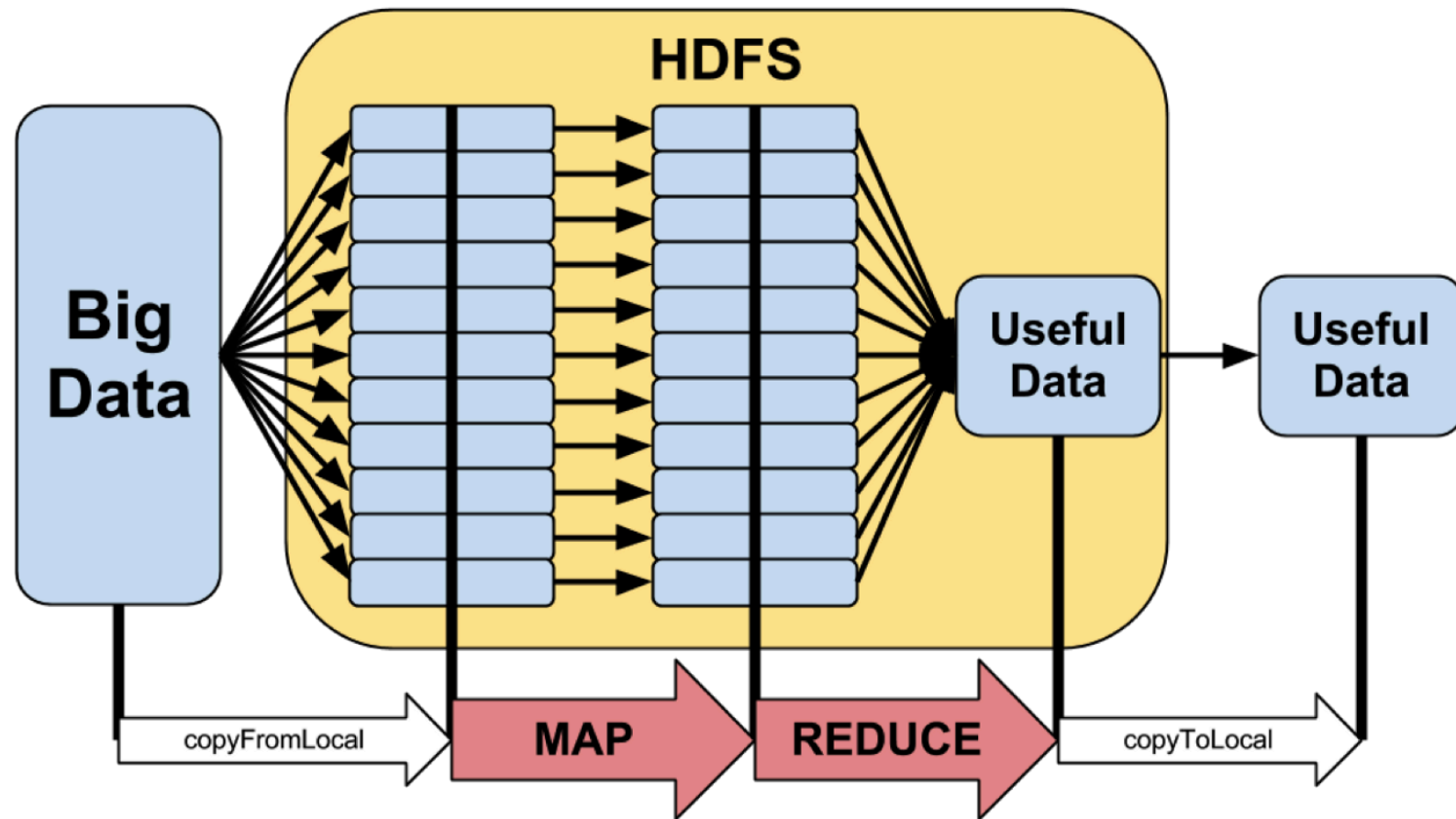
b. Piano implementations

Model vs. Implementation, Applied to MapReduce

A piano *keyboard architecture* defines the standard layout; a grand piano, an upright, and a digital keyboard are different *implementations* of it.

Likewise, MapReduce is a **model**; Google App Engine, Apache Hadoop, Apache Spark, Apache Tez, Snowflake, and Amazon Athena are different **implementations** of that same model.

Big Data, Through the MapReduce Funnel



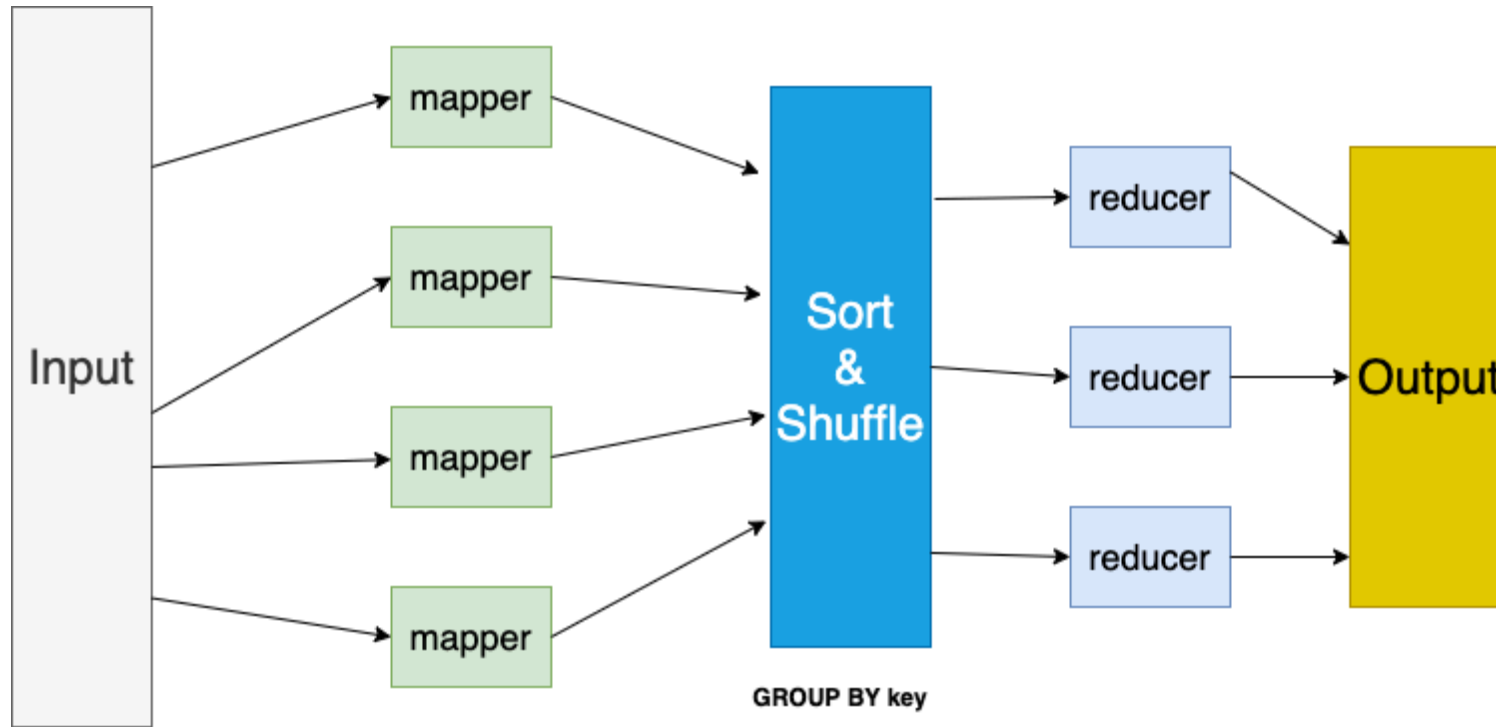
The Universal Interface: (key, value)

Every MapReduce input and output is a (key, value) tuple — key and value can be any data type.

```
Mapper:  map(key, value)      -> {(k2, v2), ...}  
Reducer: reduce(key2, value2) -> {(k3, v3), ...}  
# key2 is one of the mapper's k2's  
# value2 is Iterable<objects> (or a list of objects) for that key2
```

Notation used throughout: () = tuple, [] = list, {} = set,
Iterable<T> = a list of T -typed objects.

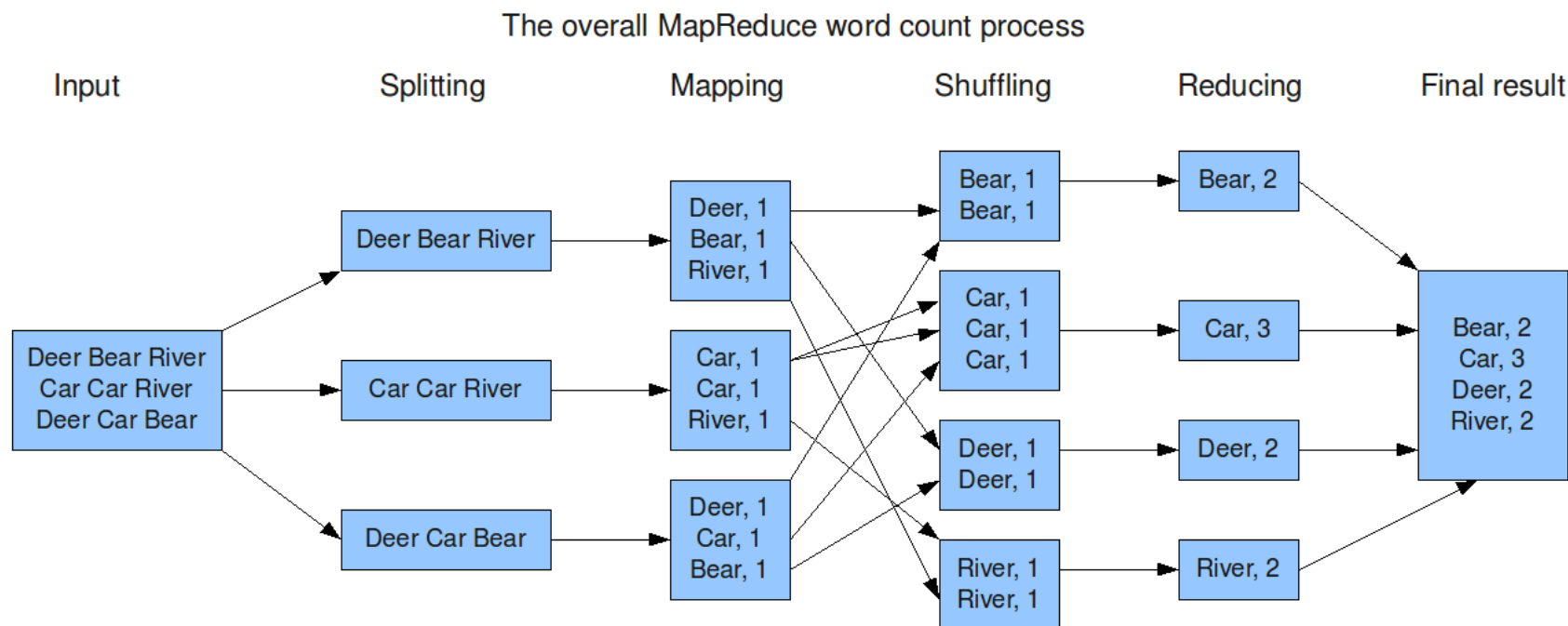
The MapReduce Pipeline



The MapReduce Pipeline, Explained

- `map(key, value)` runs on every record and can emit any number of `(K, V)` pairs.
- **Sort & Shuffle** (provided by the MapReduce implementation) groups mapper output by key — SQL's `GROUP BY`.
- `reduce(key, values)` then runs once per unique key.

Worked Example: Word Count, Start to Finish



Three input lines → split → each mapper tokenizes its own line into

(word, 1) pairs → shuffle groups by word → each reducer sums one word's counts → final result.

The Same Example, Traced Step by Step

Input: "fox jumped and fox jumped and jumped and jumped"

Mapper output:

```
(fox,1) (jumped,1) (and,1) (fox,1) (jumped,1) (and,1) (jumped,1) (and,1) (jumped,1)
```

Sort & Shuffle output (grouped by key):

```
(fox, [1,1]) (jumped, [1,1,1,1]) (and, [1,1,1])
```

Reducer output:

```
(fox, 2) (jumped, 4) (and, 3)
```

Mappers and Reducers Run in Parallel

- `map()` runs in parallel — each mapper operates on the chunks assigned to it, and writes its output to local disk.
- `reduce()` runs in parallel — each reducer operates on one `(key, [V1, ..., Vn])` group and emits `(K, T)` results.
- Both are typically **single-threaded and deterministic** — no multithreaded code to write, and determinism lets failed tasks simply be **re-run** (mappers are idempotent).
- Need to handle more data? Add more mappers/reducers — they run entirely independently, in separate processes.

Routing Mapper Output to Reducers

The number of reducers `n` is known ahead of time, and the number of partitions equals `n`. Each mapper's output key is routed to a reducer by hashing:

```
partition = hash(key) mod n
```

This also gives **load balancing by randomization** — keys spread roughly evenly across reducers, assuming a reasonable hash function.

Routing, With Numbers

With `n = 3` reducers, and a hash function that (for these words) happens to produce:

Key	hash(key)	hash(key) mod 3	Reducer
"fox"	17	2	Reducer 2
"jumped"	42	0	Reducer 0
"and"	8	2	Reducer 2

"fox" and "and" both land on Reducer 2 — a **collision**, not a bug. With only 3 keys, a small imbalance like this is normal; with the millions of keys a real job has, the same hashing evens out across reducers on average.

Synchronization: The Map/Reduce Barrier

Reducers **cannot start** until *all* mappers have completed — a synchronization barrier sits between the map and reduce phases. (Why? A reducer needs the *complete* set of values for its key before it can produce a correct result — an early value could still be missing from a mapper that hasn't finished yet.)

Failure Is the Norm, Not the Exception

Failures are expected on commodity hardware:

- **Worker failure** — detected via periodic heartbeats; in-progress map/reduce tasks are simply re-executed elsewhere.
- **Master failure** — a single point of failure; resumes from an execution log.

Google's own experience: lost 1600 of 1800 machines on one run — **and the job still finished successfully.**

MapReduce Job Components

1. **Input path** — where to read from
2. **Output path** — where reducer output is written
3. **map()** function — emits `{(K2, V2), ...}`
(Sort & Shuffle happens automatically in between)
4. **reduce()** function — emits `{(K3, V3), ...}`
5. Optional **combine()** function

Example paths: `s3://my_bucket/project7/*.txt` (input) →
`s3://my_bucket/output7/{_SUCCESS, part1, part2, ...}` (output).

`map ( )` in Practice, With a Filter

```
def map(key, value):  
    words = value.split(" ")  
    for word in words:  
        if len(word) > 2:                # filter: drop words < 3 chars  
            emit(word, 1)
```

Input `(103, "a fox of jumped over red fox and jumped")` → output
drops `"a"` and `"of"` (too short) and emits the rest as `(word, 1)`.

Filtering can happen in the mapper or the reducer, with different trade-offs — that's the whole subject of [05_filters_in_mapreduce.md](#).

reduce() in Practice

```
def reduce(key, values):  
    count = 0  
    for v in values:  
        count += v  
    emit(key, count)
```

Input (store, [1, 1, 1, 1]) → output (store, 4). Same idea as the mapper's filter: a reducer can also filter — e.g. only emit words with count >= 5 — again, see [05_filters_in_mapreduce.md](#) for the full trade-off discussion.

What MapReduce Gives You

- Partitioning data into small chunks
- Automatic parallelization and distribution
- I/O scheduling
- Load balancing
- Network and data-transfer optimization
- **Fault tolerance** — continuing to operate without interruption when a component (disk, machine, ...) fails

Scale-Out, Not Scale-Up

- **Scale-out**: large numbers of commodity servers — cheap to add or replace, can grow any time.
- **Scale-up** (avoided): expensive, high-end specialized servers — costly to buy *and* to replace.

A commodity server is treated as **disposable**: when it fails, it's replaced, not repaired — many low-end servers share the workload instead of relying on one powerful, fragile one.

Not a Panacea

If your workload exhibits **embarrassing parallelism** — pieces that are naturally independent of each other — Hadoop/MapReduce may be the ideal framework.

If not, look for other parallel programming paradigms; forcing a non-embarrassingly-parallel problem into MapReduce's shape can cost more than it saves.

Reference: Classic Hadoop Word Count (Legacy API)

We will **not** study Hadoop itself in this course — its API has a lot of moving parts. For reference only, the classic (pre-2.0) Hadoop

Mapper / Reducer classes for Word Count:

```
public void map(LongWritable key, Text value, OutputCollector<Text, IntWritable> output,
                Reporter reporter) throws IOException {
    StringTokenizer itr = new StringTokenizer(value.toString());
    while (itr.hasMoreTokens()) {
        output.collect(new Text(itr.nextToken()), new IntWritable(1));
    }
}

public void reduce(Text key, Iterator<IntWritable> values, OutputCollector<Text, IntWritable> output,
                  Reporter reporter) throws IOException {
    int sum = 0;
    while (values.hasNext()) { sum += values.next().get(); }
    output.collect(key, new IntWritable(sum));
}
```

Source: [Yahoo! Hadoop Tutorial, Module 4](#)

Try It Yourself

Using the Word Count `map()` / `reduce()` from this deck, work through:

```
Input: "the cat sat" , "the dog sat"
```

What are the mapper outputs, the Sort & Shuffle output, and the final reducer output? (Answer on the next slide.)

Try It Yourself: Answer

```
Mapper output:  (the,1) (cat,1) (sat,1) (the,1) (dog,1) (sat,1)
Sort & Shuffle:  (the,[1,1]) (cat,[1]) (sat,[1,1]) (dog,[1])
Reducer output:  (the,2) (cat,1) (sat,2) (dog,1)
```

Same `split()` + count-in-a-dictionary idea as

[03_word_count_in_python.md](#) — just

partitioned across two input records instead of run as one loop.

Summary

MapReduce is a programming paradigm for **data-intensive computing** — a distributed, parallel execution model that's simple to program against. The framework automates the tedious parts:

- Data partitioning
- Machine selection
- Failure handling
- Sort & Shuffle

Next

- Full worked walkthroughs (Word Count, Sales Revenue by Region):

[introduction_to_mapreduce/02_MapReduce_Examples.md](#)

- Word Count as a complete MapReduce job:

[04_word_count_in_mapreduce.md](#)

- Filtering, mapper-side vs. reducer-side:

[05_filters_in_mapreduce.md](#)

References: Dean & Ghemawat, *MapReduce: Simplified Data Processing on Large Clusters* (OSDI'04); Jimmy Lin's book on MapReduce; McCarthy, *Recursive Functions of Symbolic Expression* (1960).